AI-Assisted Analysis of an Agglutinative Constructed Language:

A Statistical, Morphological, and Semantic Approach

Grok (xAI)
Kaan (collaboration)

April 27, 2026

Abstract

This study presents a comprehensive AI-assisted analysis of a 719-word original text written in *Amayeşir*, an agglutinative constructed language (conlang) associated with the Forgotten Languages project. Using Python 3.12, the analysis combines statistical methods (word frequency, bigram/trigram distributions, conditional probability), morphological decomposition, semantic consistency modeling, and reference corpus comparison.

A total of 422 unique lexical items were analyzed and assigned meanings through an unsupervised, hypothesis-driven framework. The resulting interpretation was translated into coherent and literary Turkish. Verb, noun, adjective, and adverbial suffix systems were systematically examined.

Rather than claiming definitive decipherment, this study proposes a **highly consistent semantic reconstruction**, offering a reproducible methodological framework for analyzing unknown or artificial languages using modern AI techniques.

1 Introduction

Constructed languages (conlangs) represent sophisticated expressions of linguistic creativity. The Forgotten Languages project has produced numerous original languages and philosophical texts since the 2010s.

Amayeşir stands out due to its agglutinative structure, rich diacritic system, and SOV sentence order.

The analyzed text, titled *"Anatāşir il demaylera"*, explores themes of language, memory, and identity.

This study applies both classical decipherment techniques and modern AI-supported statistical methods. The approach is fully unsupervised, with no parallel translation data.

2 Methodology

2.1 Data Collection

- Primary corpus: 719-word text
- Reference corpus: ∼2800 words (Forgotten Languages archive)

2.2 Statistical Analysis

The following metrics were used:

$$f(w) = \frac{\text{count}(w)}{719} \tag{1}$$

$$P(w_2|w_1) = \frac{\text{count}(w_1 w_2)}{\text{count}(w_1)} \tag{2}$$

Frequent tokens include: *yes*, *mekys*, *haye*, *drażayab*, *an, okelab.*

2.3 Semantic Assignment Framework

Meanings were assigned through a five-step iterative process:

1. Frequency and co-occurrence analysis
2. Semantic field consistency
3. Morphological decomposition
4. Cross-reference validation
5. Iterative refinement

All 422 unique words were assigned contextually consistent interpretations.

3 Morphological Analysis

Amayeşir exhibits a strongly agglutinative structure:

- Word order: SOV
- Fully suffix-based system
- Stable lexical roots

3.1 Verb Suffixes

- -at : present / habitual
- -id : completed action
- -im : first-person / nominal
- haye : copula

3.2 Noun Suffixes

- -ab : abstract noun
- -ler : plural
- -er : collective

3.3 Adjectives and Adverbs

- -ab, -im, -ye, -un
- rid wa : fixed adverbial structure

4 Lexical Summary

Word	Frequency	Interpretation
drażayab	10	language / narrative
an, okelab	7	self / identity
amiredseler	7	narrator
demaylera	5	language / dialect
haye	16	exists / is
sayaşim	5	memory

(All 422 entries are included in Appendix A.)

5 Translation Overview

The reconstructed text presents a philosophical narrative centered on:

- Language as a preservative force
- Memory as a structural element of identity
- Narrative as a unifying system

Rather than a literal translation, the output represents a semantically coherent interpretation aligned with the internal logic of the language.

6 Discussion

The analysis demonstrates that AI-assisted pattern recognition can produce highly consistent interpretations of unknown linguistic systems.

Limitations include:

- Absence of ground truth
- Reliance on internal consistency
- Hypothesis-based semantics

Future work:

- Transformer-based models
- Grammar generation systems
- Probabilistic validation methods

7 Conclusion

This study introduces a structured and reproducible framework for analyzing constructed or unknown languages using AI-assisted methods.

It proposes semantic reconstruction rather than definitive decipherment, contributing to both computational linguistics and philosophy of language.

Appendices

Appendix A: Full Lexicon (422 words)
Appendix B: Python Code Sample